KnowSelf: Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Teaching LLM agents when to think fast, reflect, or seek external knowledge

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Current Agent Training Uses 'Flood Irrigation' — Indiscriminate Knowledge Injection Hurts Efficiency

Direct Trajectory Imitation

e.g., ReAct

Gold trajectories injected at every step regardless of need.

Trial-and-Error Refinement

e.g., Reflexion

External feedback loops applied indiscriminately.

Knowledge-Augmented Planning

e.g., KnowAgent, WKM

Domain knowledge flooded into all decision steps.

Insight: Humans possess situational self-awareness — the metacognitive ability to assess whether they can handle a situation directly, need to pause and reflect, or must seek external help. Current agents lack this entirely.

Three Situational Modes: Fast, Slow, and Knowledgeable Thinking

Agent Decision Step

Fast Thinking

Criterion: Predicted action matches gold action

Direct action with minimal deliberation

Slow Thinking

Criterion: Reflection fixes incorrect action

Self-correct through reflection

Knowledgeable Thinking

Criterion: Still incorrect after reflection

Consult external knowledge base

Special tokens mark each situation type, enabling the model to learn self-aware behavior.

KnowSelf Framework: Data Construction → Two-Stage Training → Self-Aware Inference

Step 1

Self-Awareness Data Construction

- Agent self-explores environments
- Heuristic criterion classifies each step:
  - Match gold → Fast
  - Reflection fixes → Slow
  - Otherwise → Knowledgeable
- Special tokens inserted to mark situation type

→

Step 2

Two-Stage Training

- Stage A: Supervised Fine-Tuning (SFT)
  - Teaches initial self-awareness patterns
- Stage B: Reward-based Policy Optimization (RPO)
  - Refines generation of appropriate special tokens
  - Maximizes alignment between token and true situation

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Step 3

Self-Aware Inference

- During deployment, agent generates special tokens
- Token triggers one of three paths:
  - Direct action (Fast)
  - Reflection loop (Slow)
  - Knowledge retrieval from external KB (Knowledgeable)

KnowSelf Correctly Heats a Mug and Places a Saltshaker Where Baselines Fail

Task: Put a hot mug in cabinet

OpenAI-O1:
Incorrectly tries to take an egg from the microwave after opening it.

KnowSelf (Llama-8B):
Uses a Reflection token to realize it should heat the mug using the open microwave, then proceeds correctly.

Task: Put a saltshaker in drawer

DeepSeek-R1:
Reflects that the drawer is open and tries to close it first — which fails.

KnowSelf:
Applies a Knowledge token noting that the agent should place the object without removing unrelated items, succeeding directly.

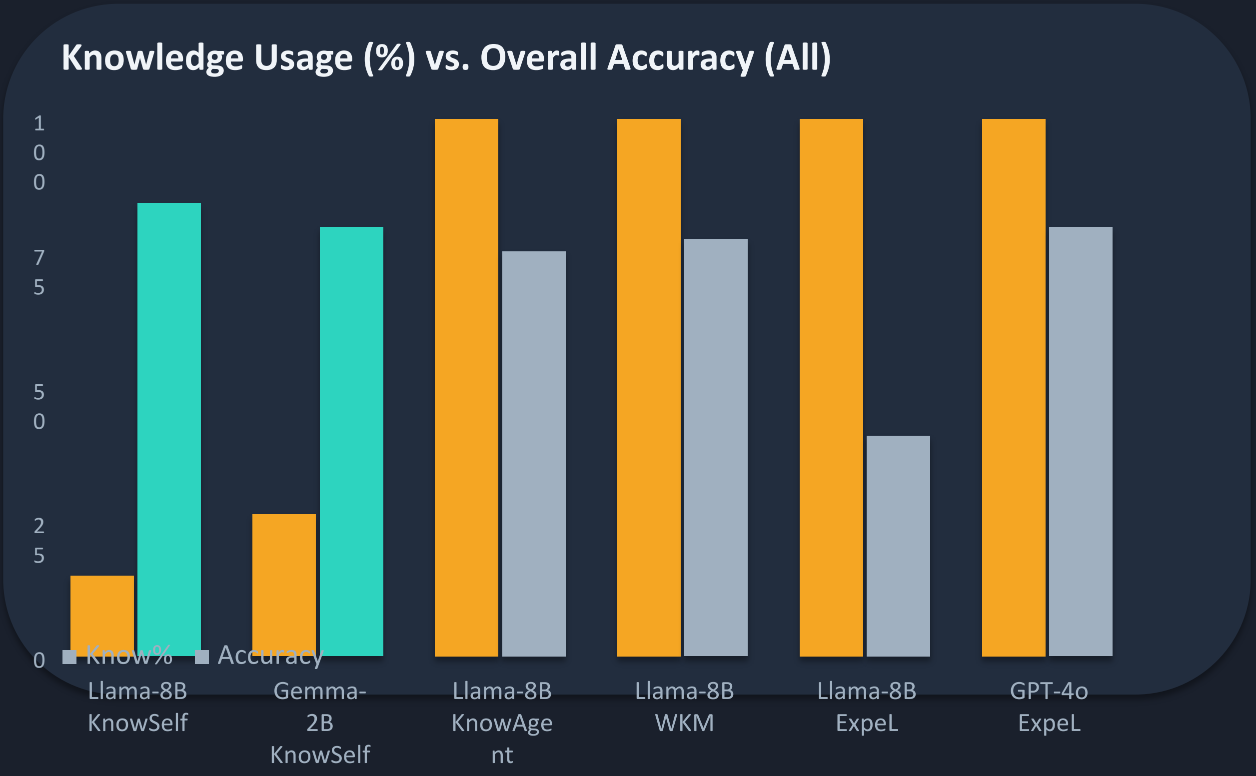
Qualitative comparison: KnowSelf vs. OpenAI-O1 and DeepSeek-R1 (see Figure 1)

KnowSelf Achieves 84.33% on ALFWorld with Llama-8B Using Only 15% Knowledge

GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	**KnowSelf**	**15.01%**	**91.67**	**87.10**	**91.30**	**85.71**	**77.78**	**64.71**	**84.33**
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	**KnowSelf**	**26.41%**	**87.50**	**93.55**	**73.91**	**76.19**	**83.33**	**52.94**	**79.85**

KnowSelf (bold, teal) uses 15–26% knowledge yet outperforms methods using 100% knowledge.

Minimal Knowledge, Maximum Impact: KnowSelf's Efficiency Advantage Over Full-Knowledge Methods



- KnowSelf (Llama-8B) achieves 84.33% overall — surpassing GPT-4o ExpeL (79.85%) — while using only 15% knowledge.
- Even Gemma-2B reaches 79.85% with KnowSelf, matching GPT-4o ExpeL and exceeding all other Gemma-2B baselines.
- Methods using 100% knowledge (ExpeL, KnowAgent, WKM) often underperform KnowSelf, confirming indiscriminate injection hurts performance.
- The special-token mechanism is lightweight and data-efficient, requiring only heuristic labeling on self-explored trajectories.

Key Takeaways: Self-Awareness Is the Missing Piece in Agent Planning

1. **Indiscriminate knowledge injection is wasteful and suboptimal**

   Current methods flood agents with gold trajectories, feedback, or domain knowledge regardless of need — leading to fragile planning and high inference costs.

2. **Agents can learn situational self-awareness via special tokens**

   Lightweight heuristic labeling on self-explored trajectories is sufficient; no expensive human annotation or massive external feedback required.

3. **KnowSelf (Llama-8B) outperforms strong baselines including GPT-4o**

   84.33% vs. GPT-4o ReAct (76.12%) and GPT-4o ExpeL (79.85%) — selective knowledge use beats blanket application.

4. **Only 15–26% knowledge usage needed versus 100% for competing methods**

   KnowSelf queries external knowledge sparingly, yet achieves the best overall scores on ALFWorld across task types.

5. **The paradigm generalizes across model scales**

   Even Gemma-2B reaches 79.85%, matching GPT-4o ExpeL. The three-situation taxonomy provides a principled framework for autonomous cognitive regulation.

Future Directions: Extend to multi-agent coordination, dynamic knowledge-base updates, and real-world robotic planning.